

2023



Migrating a Mission-Critical SQL Server Workload to Amazon EBS io2 Block Express with Cvent

Learn how Cvent, a leading provider of events and hospitality technology, migrated its critical SQL Server workload to Amazon EBS io2 Block Express.

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66%

reduced service interruptions during u



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up to 64 TB

increased storage capacity and improv



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Overview

[Cvent](#), one of the largest event and hospitality technology companies in the world, completed a major initiative in 2021 to retire its on-premises data centers and migrate all its workloads to Amazon Web Services (AWS). As part of this migration, it needed to migrate its mission-critical, highly transactional SQL Server workload, which supports Cvent's Event and Hospitality Cloud platform, to the cloud. Downtime for the SQL Server workload would have a direct impact on Cvent's customers. By migrating its critical SQL Server workload to AWS, Cvent increased the speed and flexibility of upgrades, reduced service interruptions during upgrades, and improved its failover recovery, scalability, and cost management.



Opportunity | Using Amazon EBS for Migration of a Mission-Critical Workload for Cvent

Founded in 1999, Cvent provides software solutions to the entire meetings and events system to streamline processes, maximize engagement, and deliver more impactful experiences among other tasks. In 2015, the company began exploring new data center opportunities to better support its growing international customer base while improving security. To do this, Cvent used AWS services to create smaller workloads with lower traffic requirements. "Building on AWS in Europe gave us the confidence that we could run all our applications and services for North America on AWS as well," says Mike Tiffany, senior director of database engineering.

Deciding to retire its on-premises data centers in North America, Cvent first migrated its small workloads to AWS. The final piece of Cvent's North America migration to AWS was its SQL Server workload—its extremely critical

component that involved tens of terabytes of data and had significant storage performance requirements. The SQL Server workload was highly transactional and posed a great risk to migration. Cvent first chose to use [Amazon FSx for Windows File Server](#), which provides fully managed shared storage built on Windows Server and delivers a wide range of data access, data management, and administrative capabilities.

In the summer of 2022, a new instance type that supported Cvent’s CPU and memory requirements became available on AWS. So the company rearchitected its SQL Server workload to run on [Amazon Elastic Block Store](#) (Amazon EBS), a scalable, high-performance block storage service. It migrated its SQL Server from a failover cluster instance architecture to Always On availability groups in 3 months. Using [Amazon EBS io2 Block Express volumes](#), which are designed to deliver sub-millisecond latency and 99.999 percent durability, Cvent keeps three independent copies of its database in sync across three different AWS Availability Zones. By maintaining copies across AWS Availability Zones, the company is following guidelines from [AWS Well-Architected Framework](#), which provides learning architectural best practices for designing and operating reliable, secure, efficient, cost-effective, and sustainable systems in the cloud. After completing this two-step migration, Cvent retired its on-premises data centers. “Collaboration with the AWS team throughout the summer was extensive, and AWS support was invaluable,” says Tiffany.

Solution | Reducing Service Interruptions during Upgrades by 66% and Improving Reliability and Performance Using AWS

One benefit of the migration to AWS was improved resilience. Using Amazon EBS io2 Block Express volumes and SQL Server Always On availability groups, Cvent meets its resilience requirements even if one of its three node clusters, each in a different Availability Zone, goes down. This is an improvement over the on-premises infrastructure, where all resources were hosted in the same data center. With this architecture, Cvent enhanced its failover recovery speed by 80–90 percent. Additionally, using [AWS Backup](#), a cost-effective, fully managed, policy-based service that simplifies data protection at scale, Cvent can take snapshots of database instances to quickly restore a node when needed.

Moreover, using Amazon EBS io2 Block Express volumes, Cvent improved performance, delivering higher IOPS that exceeded the performance of its on-premises solution. Cvent used the volume capabilities of [Amazon EBS io2 Block Express](#) to scale IOPS and used Amazon EBS Snapshots to create consistent backups of Amazon EBS volumes.



In September 2023, 1 year after migrating to Amazon EBS, Cvent performed the first major version upgrade since migrating its SQL Server to AWS. This upgrade involved two of its three SQL Server clusters, totaling 350 TB of Amazon EBS io2 Block Express storage. “When we’d done SQL Server upgrades in production in the past, we’d expected some service disruption and customer impact,” says Tiffany. “Using this new architecture based on Amazon EBS io2 Block Express and Always On availability groups, we reduced the impact to customers by 66 percent.”

The migration to AWS and the re-architecture of SQL Server have resulted in substantial productivity benefits for Cvent. On premises, an SQL Server hardware refresh was measured in months because of hardware evaluation and procurement cycles. By taking advantage of infrastructure as code in AWS, the company can spin up new servers and clusters with a minor code change and deployment, saving a significant amount of time and effort. Cvent has used these capabilities to test and deploy new instance types and optimized for cost by deploying new instance and storage configurations much faster than they could have in their on-premise data centers. “By using AWS, we can innovate quickly by focusing on solving the issue instead of trying to get the plumbing behind it done correctly,” says Kunal Jain, senior architect of database engineering at Cvent.

Using Amazon EBS io2 Block Express volumes, Cvent has more room to scale and has doubled its throughput capacity. Cvent uses [Amazon Elastic Compute Cloud](#) (Amazon EC2), secure and resizable compute capacity for virtually any workload, and can change the size of its [Amazon EC2 instances](#) and modify storage to manage costs on the basis of use.

“Using AWS, we now have more control and flexibility, so we can focus more on efforts that add value for our customers. We’re in a better place overall,” says Tiffany.

Outcome | Going Serverless and Modernizing Applications on AWS

Cvent’s migration to AWS started as a lift and shift, followed by the company modernizing and taking advantage of managed services. Cvent’s next goal is to go serverless wherever possible. The company has active research and development projects and intends to start going serverless in 2024.

“We have a lot of other workloads using AWS technologies,” says Tiffany. “Our goal is always to match each workload to the right technology.”

About Cvent

Cvent is a leading provider of meetings, events, and hospitality technology with over 21,000 customers worldwide. Cvent solutions optimize the event management value chain and have helped clients to manage millions of meetings and events.

AWS Services Used



Amazon EBS io2 Block Express Volumes

Amazon Elastic Block Store (Amazon EBS) is an easy-to-use, scalable, high-performance block-storage service designed for Amazon Elastic Compute Cloud (Amazon EC2). EBS io2 Block Express volumes are designed to deliver per volume performance of up to 4,000 MB/s throughput, 256,000 IOPS, 64 TiB storage capacity, and 1,000 IOPS/GB, as well as 99.999% durability and sub-millisecond latency.

[Learn more »](#)

Amazon FSx for Windows File Server

Amazon FSx for Windows File Server provides fully managed shared storage built on Windows Server, and delivers a wide range of data access, data management, and administrative capabilities.

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AWS Backup

AWS Backup is a cost-effective, fully managed, policy-based service that simplifies data protection at scale.

[Learn more »](#)

Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 700 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

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